AVEVA™ E3D Design

3.1

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

No part of this publication may be reproduced, stored in a retrieval system, or transmitted, in any form or by any means, mechanical, photocopying, recording, or otherwise, without the prior written permission of AVEVA Group Limited. No liability is assumed with respect to the use of the information contained herein.

Although precaution has been taken in the preparation of this documentation, AVEVA assumes no responsibility for errors or omissions. The information in this documentation is subject to change without notice and does not represent a commitment on the part of AVEVA. The software described in this documentation is furnished under a license agreement. This software may be used or copied only in accordance with the terms of such license agreement. AVEVA, the AVEVA logo and logotype, OSIsoft, the OSIsoft logo and logotype, ArchestrA, Avantis, Citect, DYNSIM, eDNA, EYESIM, InBatch, InduSoft, InStep, IntelaTrac, InTouch, Managed PI, OASyS, OSIsoft Advanced Services, OSIsoft Cloud Services, OSIsoft Connected Services, OSIsoft EDS, PIPEPHASE, PI ACE, PI Advanced Computing Engine, PI AF SDK, PI API, PI Asset Framework, PI Audit Viewer, PI Builder, PI Cloud Connect, PI Connectors, PI Data Archive, PI DataLink, PI DataLink Server, PI Developers Club, PI Integrator for Business Analytics, PI Interfaces, PI JDBC Driver, PI Manual Logger, PI Notifications, PI ODBC Driver, PI OLEDB Enterprise, PI OLEDB Provider, PI OPC DA Server, PI OPC HDA Server, PI ProcessBook, PI SDK, PI Server, PI Square, PI System, PI System Access, PI Vision, PI Visualization Suite, PI Web API, PI WebParts, PI Web Services, PRiSM, PRO/II, PROVISION, ROMeo, RLINK, RtReports, SIM4ME, SimCentral, SimSci, Skelta, SmartGlance, Spiral Software, WindowMaker, WindowViewer, and Wonderware are trademarks of AVEVA and/or its subsidiaries. All other brands may be trademarks of their respective owners.

U.S. GOVERNMENT RIGHTS

Use, duplication or disclosure by the U.S. Government is subject to restrictions set forth in the license agreement with AVEVA Group Limited or its subsidiaries and as provided in DFARS 227.7202, DFARS 252.227-7013, FAR 12-212, FAR 52.227-19, or their successors, as applicable.

AVEVA Legal Resources: https://www.aveva.com/en/legal/

AVEVA Third Party Software Notices and Licenses: https://www.aveva.com/en/legal/third-party-software-license/

Contents

XML Schema 4
- System Architecture 4
- Basic Concepts 5
- Annotated .xml Output File from PSI 8
- Restraint Information 19
- Return Process 21

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

XML Schema

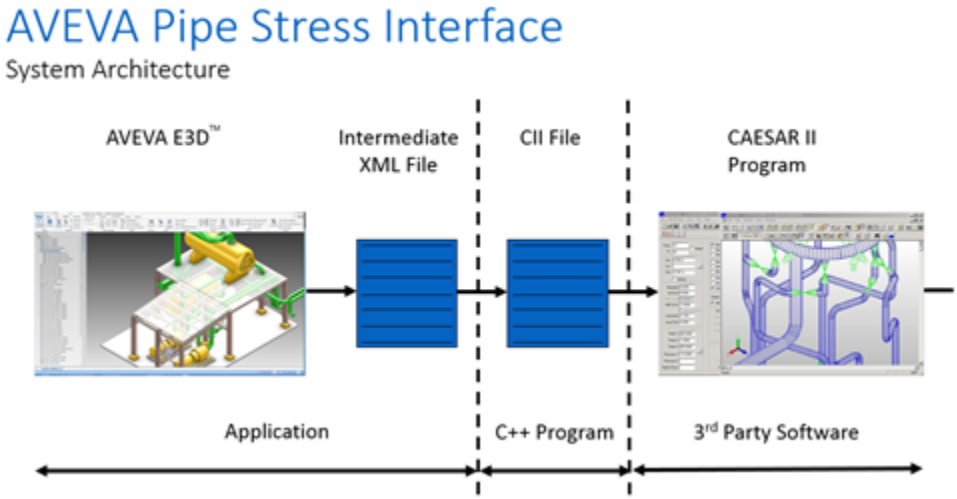
The AVEVA Pipe Stress Interface Package (PSI) creates an xml file which is converted into data for one or more external pipe stress software systems. The default software system is CAESAR but it also supports other software systems that support the CII file format. The XML data represents the AVEVA E3D Design model in the form of a series of nodes which represent the piping route and all of its components.

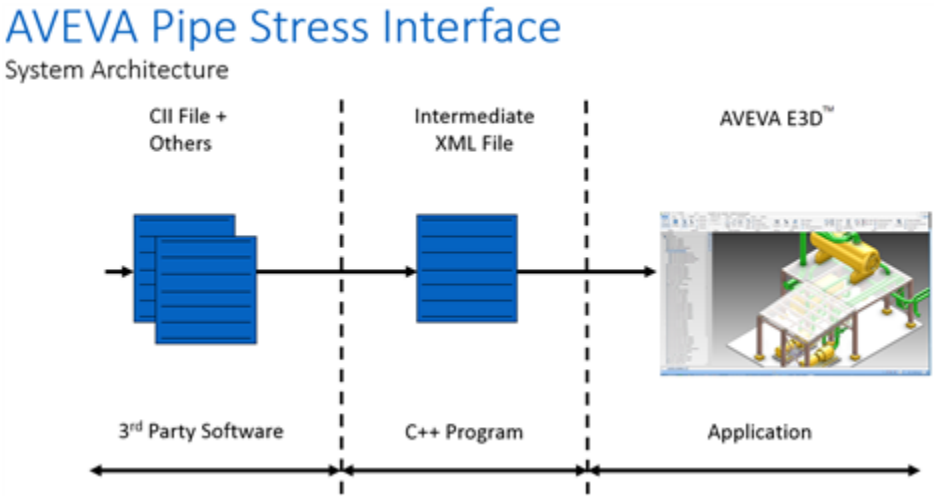
The PSI reads the XML file and produces a CII file for further processing. The following sections describe the format of the XML file and how it represents the AVEVA E3D Design model.

System Architecture

The image below shows the default system architecture for the PSI software. If a software vendor requires to write their own adapters to their software the below architecture should be considered.

The first step will be to write a program to convert the XML output to the software vendor's format. The second step will be to write a program to convert the output from the software vendor's program to the XML format required by the PSI software.





© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

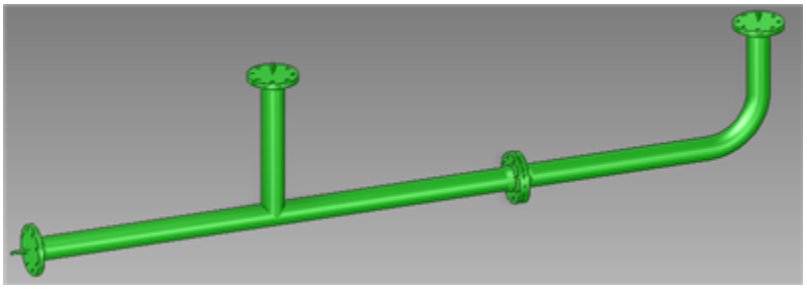
Basic Concepts

The PSI parsing mechanism steps through a series of branches in the sequence of the stress pipe.

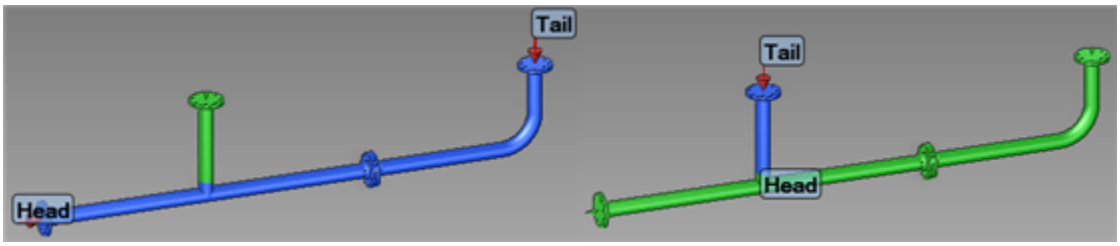
Note:
A stress pipe is a collection of branches which do not need to belong to the same real pipe.

The AVEVA E3D Design pipe structure is as follows:

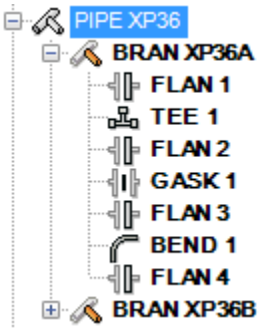
A pipe is a collection of two ended segments called branches. In the example below there are two branches, the horizontal segment including the tee, bend and two flanges. The second branch is from the tee to the flange. A pipe may contain multiple branches.



Branches have a start and end point known as the head and tail. These define position, direction and many other attributes that define the branch.



The simplest form of a branch is one which runs directly from head to tail with no components other than the tube between them. Most branches do have components and these are contained in a list which exists below the branch, as shown below:



A branch can be summarized as a two ended container for components.The only important rule is that the components must be in sequence, in line and face in the correct direction. Individual branch components are defined in a piping catalogue and selected from a pipe specification. The list of components may be any of the following:

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

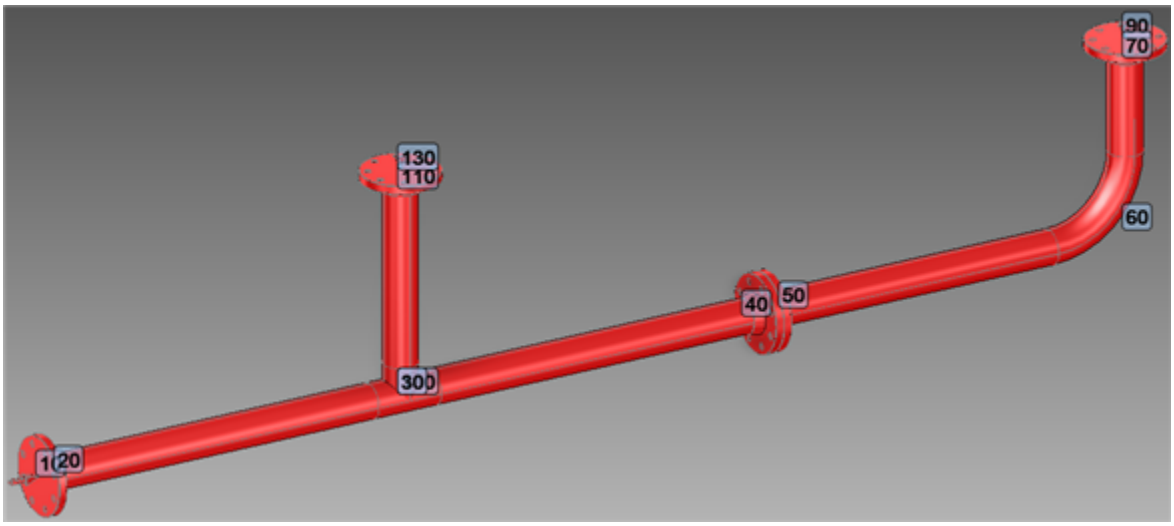
FLAN	Flange
ELBO	Elbow
TEE	Tee
REDU	Reducer
VALV	Valve
ATTA	Attachment Point
PCOM	Piping Component
FBLI	Blind Flange
BEND	Bend
OLET	Olet
COUP	Coupling
INST	Instrument
CAP	Cap
GASK	Gasket
FILT	Filter
WELD	Weld
CROS	Cross
UNIO	Union
VTWA	Three Way Valve
CLOS	Closure
TRAP	Trap
LJSE	Lap Joint Stub End
FTUB	Fixed Tube
VFWA	Four Way Valve
SHU	Standard Hook Up

Some of the listed types are obvious but others are less so, this not important for the xml description.

Components within a branch must have at least two connection points, called PPoints which define the component length and position. A tee and other multi-way components with other connection points which are connected to other branches.

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

The following is the pipe represented as a stress group showing the node numbers which appear in CAESAR.



What Goes in the .xml file

The body of the xml file is essentially a list. Every PPoint is listed, but only those with Node numbers are transferred to the CII file. The output contains other values but this list contains the essential attributes required to understand the concept.

Seq	NodeNo	Type	RefNo	Point	PPoint	Bore	Radius	Rigid	East	North	Up
1		BRAN	=676/3060	1	1	80mm	0.000	START	-2000mm	13800mm	0.000mm
2	10	FLAN	=676/3061	1	1	80mm	0.000		-2000mm	13800mm	0.000mm
3	20	FLAN	=676/3061	2	2	80mm	0.000	END	-2050mm	13800mm	0.000mm
4		TEE	=676/3062	1	1	80mm	0.000		-2913mm	13800mm	0.000mm
5		TEE	=676/3062	3	3	80mm	0.000		-2999mm	13800mm	86mm
6	30	TEE	=676/3062	0	0	80mm	0.000		-2999mm	13800mm	0.00mm
7		TEE	=676/3062	2	2	80mm	0.000		-3085mm	13800mm	0.00mm
8	40	FLAN	=676/3063	1	2	80mm	0.000	START	-3949mm	13800mm	0.0mm
9		FLAN	=676/3063	2	1	80mm	0.000		-3999mm	13800mm	0.0mm

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

10		GASK	=676/3064	1	1	80mm	0.000		-3999 mm	13800 mm	0.0mm
11		GASK	=676/3064	2	2	80mm	0.000		-4001 mm	13800 mm	0.0mm
12		FLAN	=676/3065	1	1	80mm	0.000		-4001 mm	13800 mm	0.0mm
13	50	FLAN	=676/3065	2	2	80mm	0.000	END	-4051 mm	13800 mm	0.0mm
14		BEND	=676/3066	1	1	80mm	210.00		-4790 mm	13800 mm	0.0mm
15	60	BEND	=676/3066	0	0	80mm	210.00		-5000 mm	13800 mm	0.0mm
16		BEND	=676/3066	2	2	80mm	210.00		-5000 mm	13800 mm	210mm
17	70	FLAN	=676/3067	1	2	80mm	0.000	START	-5000 mm	13800 mm	450mm
18		FLAN	=676/3067	2	1	80mm	0.000		-5000 mm	13800 mm	500mm
19	90	BRAN	=676/3060	2	2	80mm	0.000	END	-5000 mm	13800 mm	500mm
20	100	BRAN	=676/3068	1	1	80mm	0.000		-2999 mm	13800 mm	0.00mm
21	110	FLAN	=676/3069	1	2	80mm	0.000	START	-2999 mm	13800 mm	536mm
22		FLAN	=676/3069	2	1	80mm	0.000		-2999 mm	13800 mm	586mm
23	130	BRAN	=676/3068	2	2	80mm	0.000	END	-2999 mm	13800 mm	586mm

Annotated .xml Output File from PSI

This is an example of an XML file written by the PSI application - the XML file used on the return path from CAESAR II. Refer to Return Process for further information.

```
<?xml version="1.0" encoding="utf-8"?>
```

Standard header information

```
<PipeStressExport xmlns="http://aveva.com/pipestress116.xsd">
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<DateTime>15:43:24 19 July 2016</DateTime>
```

The date and time at which the XML files was generated. This, and the next few elements are put into the 'title page' of the .CII file

```
<Source>AVEVA PSI</Source>
<VersionNumber>12.1.1</Version>
<UserName>Tim.Sharrock</UserName>
<Purpose>Preliminary stress run</Purpose>
<ProjectName>APS</ProjectName>
<MDBName>/ALL</MDBName>
<TitleLine>PSI stress Output</TitleLine>
<!-- Configuration information -->
<Units>
<DistanceUnits>MM</DistanceUnits>
```

The AVEVA E3D Design Distance units: which can be one of:

```
NameValue linearConversionFactors[ ]=
{ // store as mm
{"mm",1.0},
{"Inch",25.4},
{"Finch",25.4}, // assume same as inch...
{0,0.0}
};
<BoreUnits>MM</BoreUnits>
```

The AVEVA E3D Design Bore units: which can be one of:

```
NameValue linearConversionFactors[ ] =
{ // store as mm
{"mm",1.0},
{"Inch",25.4},
{"Finch",25.4}, // assume same as inch...
{0,0.0}
}
<PressureUnits>BAR G</PressureUnits>
```

The units used in the database for pressures: which can be one of:

```
NameValue pressureConversionFactors[ ] =
{ // store as Bar
{"lb./sq.in",0.068946},
{"lb./sq.ft",4.787917e-4},
{"BAR", 1.0},
{"N./sq.cm", 0.1},
{"N./sq.mm", 10.0},
{"dyne/cm2", 0.000001},
{"kpa", 0.01},
{0,0.0}
};
<TemperatureUnits>C</TemperatureUnits>
```

The units used in the database for temperatures: which can be one of:

```
NameValue temperatureConversionFactors[ ] =
{ // store as Celsius
{"C",1.0},
{"F",1.0/1.8},
{"K",1.0},
{0,0.0}
};
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
NameValue temperatureConversionOffsets[ ] =
{ // store as Celsius
{"C",0.0},
{"F",-32.0},
{"F",+273.15},
{0,0.0}
};
<WeightUnits>kg</WeightUnits>
```

The units used in the database for weights: which can be one of:

```
NameValue weightConversionfactors[] =
{ // store as kg weight
{"lb",0.45359237},
{"kg",1.0},
{"N",0.10197 },
{0,0.0}
};
<ForceUnits>KN.</ForceUnits>
```

The units used in the database for forces: which can be one of:

```
NameValue forceConversionfactors[] =
{ // store as Newtons
{"lb",4.4480},
{"kip",4448.0}, // kiloPounds...
{"N.",1.0},
{"N",1.0},
{"KN.",1000.0},
{"KN",1000.0},
{0,0.0}
};
<WallThicknessUnits>mm</WallThicknessUnits>
```

The units used in the database for wall thicknesses: which can be one of:

```
NameValue linearConversionFactors[] =
{ // store as mm
{"mm",1.0},
{"Inch",25.4},
{"Finch",25.4}, // assume same as inch...
{0,0.0}
};
<FluidDensityUnits>kg./cu.m</FluidDensityUnits>
```

The units used in the database for fluid density: which can be one of:

```
NameValue densityConversionfactors[] =
{ // store as kg./cu.m
{"lb./cu.in",27680.0},
{"lb./cu.ft",16.0185},
{"kg./cu.m", 1.0},
{"kg./cu.dm",1.0e3},
{"kg./cu.cm",1.0e6},
{"kg./cu.mm",1.0e9},
{0,0.0}
};
</Units>
<Pipe>
```

This pipe element is the "StressGroup" in PSI - it is stored as a PIPE element

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<FullName>/XP36-GROUP</FullName>
```

The name of the stress-group

```
<Ref>=2013286672/1995</Ref>
```

The AVEVA E3D Design reference of the stress-group - this is used, on the return for identifying the stress-group

```
<Branch>
```

The branch however is a "real" AVEVA E3D Design branch included in the Stress-Group

```
<Branchname>/XP36A</Branchname>
```

The name of the AVEVA E3D Design branch

```
<Temperature>
```

CAESAR II allows up to 10 temperatures at each node, AVEVA E3D Design stores one at a branch level, PSI allows you to specify extra temperatures, but not varying over the model.

```
<Temperature1>330</Temperature1>
```

All the temperature elements use the units defined in the <TemperatureUnits> element

```
<Temperature2>-100000</Temperature2>
```

-10000 is used by AVEVA E3D Design for unset values

```
<Temperature3>-100000</Temperature3>
<Temperature4>-100000</Temperature4>
<Temperature5>-100000</Temperature5>
<Temperature6>-100000</Temperature6>
<Temperature7>-100000</Temperature7>
<Temperature8>-100000</Temperature8>
<Temperature9>-100000</Temperature9>
</Temperature>
<Pressure>
```

CAESAR II allows up to 10 pressures at each node, AVEVA E3D Design stores one at a branch level, PSI allows you to specify extra pressures, but not varying over the model

```
<Pressure1>17</Pressure1>
```

All the temperature elements use the units defined in the <TemperatureUnits> element

```
<Pressure2>0</Pressure2>
```

-10000 is used by AVEVA E3D Design for unset values

```
<Pressure3>0</Pressure3>
<Pressure4>0</Pressure4>
<Pressure5>0</Pressure5>
<Pressure6>0</Pressure6>
<Pressure7>0</Pressure7>
<Pressure8>0</Pressure8>
<Pressure9>0</Pressure9>
</Pressure>
<MaterialNumber>102</MaterialNumber>
```

The material number is an integer, and is passed straight through to the .CII file.

```
<InsulationDensity>100</InsulationDensity>
<FluidDensity>1000</FluidDensity>
<Node>
<NodeNumber>-1</NodeNumber>
```

If the node-number is -1, then the node should not be represented in the output file.

```
<Endpoint>1</Endpoint>
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

The Endpoint is broadly equivalent to the AVEVA E3D Design ppoint of the node (except that endpoints 1 and 2 are the parrive and pleave, rather than ppoints 1 and 2.

For some component types we only consider some of the endpoints:

we need the node at endpoint 0 for a bend

node 0 of a tee is needed for connecting branches

when we reach endpoint 2 of a reducer we have both sets of diameters available

and there is some logic involved in identifying rigid components. What I currently use is:

```
if((componentType=="RIGID") ||
((endpoint==2) &&
((componentType=="GASK")||
(componentType=="FLAN"))||
(((endpoint==2)||(endpoint==0)) &&
(componentType=="VALV")||
(componentType=="FILT")||
(componentType=="INST")||
(componentType=="PCOM")||
(componentType=="TRAP"))))
{
data.rigidData().m_enabled=true;
```

the (componentType=="RIGID") case does not, I think occur in any XML files written by the released PSI - but did in some pre-release versions.

```
<Rigid>1</Rigid>
<ComponentType>BRAN</ComponentType>
```

The component type is the AVEVA E3D Design element type

```
<Weight>0</Weight>
```

The weight is in <WeightUnits>. For a rigid block made up of several AVEVA E3D Design components the PML will assemble, and if necessary distribute the total weight, and will assign the combined weight to suitable nodes in the XML file.

```
<ComponentRefNo>=2013286676/3060</ComponentRefNo>
```

The AVEVA E3D Design element reference - primarily used for identifying components whn data is being translated back from the stressing package.

```
<OutsideDiameter>88.9000015258789</OutsideDiameter>
```

In <BoreUnits>

```
<WallThickness>3.17</WallThickness>
```

In <WallThicknessUnits>

```
<CorrosionAllowance>1</CorrosionAllowance>
```

In <CorrosionAllowanceUnits>

```
<InsulationThickness>150</InsulationThickness>
```

In <BoreUnits>

```
<Position>6867 3700 1490</Position>
```

In <DistanceUnits> with respect to the world

```
<BendRadius>0</BendRadius>
```

In <DistanceUnits> - set to zero for "not a bend"

```
<SIF>0</SIF>
```

This element is misnamed - it does not contain a SIF value, but may contain a tee-type code as used in CAESAR II

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

(this ends up in the SIF & TEES table in the .cii file) - we may need to consider changes in this area, depending on how other stressing packages handle tees.

```
</Node>
```

We have now reached the end of the first Node element, so comments will become sparser,

```
<Node>
<NodeNumber>10</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3061</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2000.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>20</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<Rigid>2</Rigid>
<ComponentType>RIGID</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3061</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2050.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>TEE</ComponentType>
<Weight>1.31</Weight>
<ComponentRefNo>=2013286676/3062</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2913.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>3</Endpoint>
<ComponentType>TEE</ComponentType>
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<Weight>1.31</Weight>
<ComponentRefNo>=2013286676/3062</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 86.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>30</NodeNumber>
<NodeName></NodeName>
<Endpoint>0</Endpoint>
<ComponentType>TEE</ComponentType>
<Weight>1.31</Weight>
<ComponentRefNo>=2013286676/3062</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<ComponentType>TEE</ComponentType>
<Weight>1.31</Weight>
<ComponentRefNo>=2013286676/3062</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-3085.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>40</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<Rigid>1</Rigid>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3063</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-3949.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
```


© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3063</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-3999.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>GASK</ComponentType>
<Weight>0.19</Weight>
<ComponentRefNo>=2013286676/3064</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-3999.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<ComponentType>GASK</ComponentType>
<Weight>0.19</Weight>
<ComponentRefNo>=2013286676/3064</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-4001.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3065</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
```


© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<Position>-4001.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>50</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<Rigid>2</Rigid>
<ComponentType>RIGID</ComponentType>
<Weight>9.32</Weight>
<ComponentRefNo>=2013286676/3065</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-4051.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>BEND</ComponentType>
<Weight>0</Weight>
<ComponentRefNo>=2013286676/3066</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-4790.00 13800.00 0.00</Position>
<BendRadius>210</BendRadius>
<BendType>0</BendType>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>60</NodeNumber>
<NodeName></NodeName>
<Endpoint>0</Endpoint>
<ComponentType>BEND</ComponentType>
<Weight>0</Weight>
<ComponentRefNo>=2013286676/3066</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>0</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-5000.00 13800.00 0.00</Position>
<BendRadius>210</BendRadius>
<BendType>0</BendType>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
```


© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<ComponentType>BEND</ComponentType>
<Weight>0</Weight>
<ComponentRefNo>=2013286676/3066</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-5000.00 13800.00 210.00</Position>
<BendRadius>210</BendRadius>
<BendType>0</BendType>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>70</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<Rigid>1</Rigid>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3067</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-5000.00 13800.00 450.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3067</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-5000.00 13800.00 500.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>90</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<Rigid>2</Rigid>
<ComponentType>RIGID</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3060</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-5000.00 13800.00 500.00</Position>
```


© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
</Branch>
<Branch>
<Branchname>/XP36B</Branchname>
<Temperature>
<Temperature1>-100000</Temperature1>
<Temperature2>-100000</Temperature2>
<Temperature3>-100000</Temperature3>
<Temperature4>-100000</Temperature4>
<Temperature5>-100000</Temperature5>
<Temperature6>-100000</Temperature6>
<Temperature7>-100000</Temperature7>
<Temperature8>-100000</Temperature8>
<Temperature9>-100000</Temperature9>
</Temperature>
<Pressure>
<Pressure1>0</Pressure1>
<Pressure2>0</Pressure2>
<Pressure3>0</Pressure3>
<Pressure4>0</Pressure4>
<Pressure5>0</Pressure5>
<Pressure6>0</Pressure6>
<Pressure7>0</Pressure7>
<Pressure8>0</Pressure8>
<Pressure9>0</Pressure9>
</Pressure>
<MaterialNumber>0</MaterialNumber>
<InsulationDensity>100</InsulationDensity>
<FluidDensity>0</FluidDensity>
<Node>
<NodeNumber>100</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>BRAN</ComponentType>
<Weight>0</Weight>
<ComponentRefNo>=2013286676/3068</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>110</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<Rigid>1</Rigid>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3069</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
```

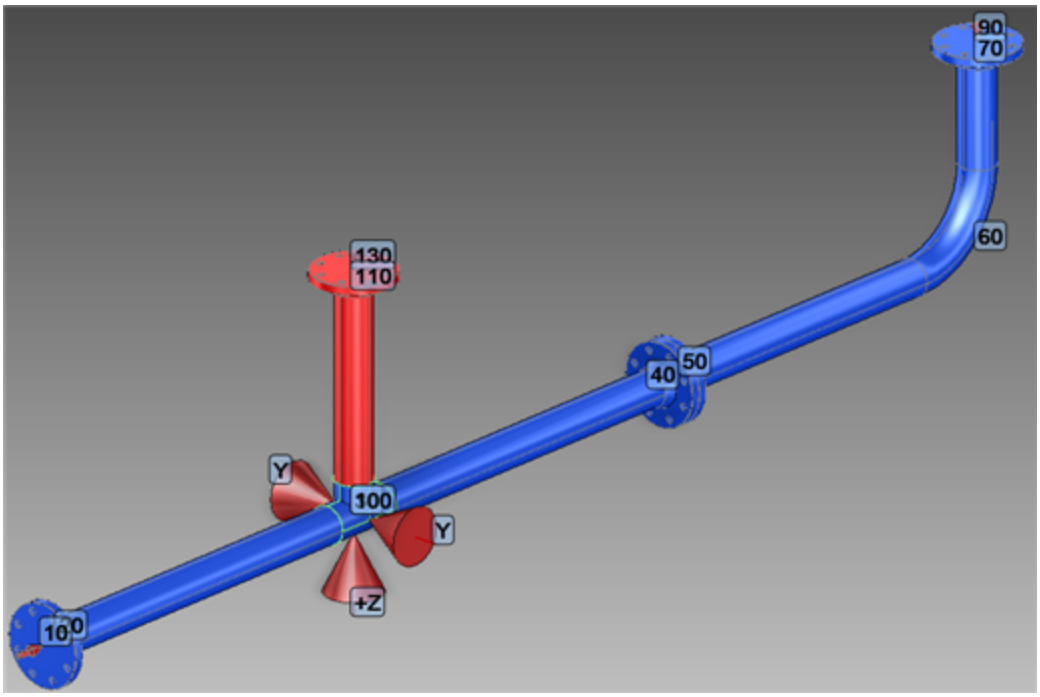

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 536.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>-1</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3069</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 586.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
<Node>
<NodeNumber>130</NodeNumber>
<NodeName></NodeName>
<Endpoint>2</Endpoint>
<Rigid>2</Rigid>
<ComponentType>RIGID</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=2013286676/3068</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 586.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
</Node>
</Branch>
</Pipe>
</PipeStressExport>
```


Restraint Information

All nodes can contain restraint information in the form of up to four additional restraints.

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.



Here we have added two restraints +Z and Y. This changes the xml output as follows:

```
<Node>
<NodeNumber>30</NodeNumber>
<NodeName></NodeName>
<Endpoint>0</Endpoint>
<ComponentType>TEE</ComponentType>
<Weight>1.31</Weight>
<ComponentRefNo>=2013286676/3062</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2999.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
```

For each component containing a restraint, additional restraint blocks are added to the node which contains the node number. Here we have two restraints

```
<Restraint>
<Type>+Z</Type>
<Stiffness>0</Stiffness>
<Gap>0</Gap>
<Friction>0</Friction>
</Restraint>
<Restraint>
<Type>Y</Type>
<Stiffness>0</Stiffness>
<Gap>0</Gap>
<Friction>0</Friction>
</Restraint>
</Node>
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

Return Process

After stressing we produce two output files from CAESAR and these are combined to form a return xml file.

The first is a new CII file containing any dimensional changes and possible new elements. This is a CII file produced from the CAESAR model after stressing.

The second file is a list of deflections per node from a CAESAR load case report.

To produce a return input to the 3D model, these two files are combined with the original xml output to form a new xml file containing the stress data. In essence the xml format is the same as the output but with new information appended.

The first new item is the load case List which then contains a list of individual load case names. In the following examples the additional lines are shown in Blue:

```
<?xml version="1.0" encoding="utf-8"?>
<PipeStressExport xmlns="http://aveva.com/pipeStress116.xsd">
<LoadCaseList>
<LoadCaseNames>CASE 1 (SUS) W</LoadCaseNames>
…..
</LoadCaseList>
```

The sequence then continues replicating the individual components exactly as they were in the original output but with extra blocks.

```
<NodeNumber>10</NodeNumber>
<NodeName></NodeName>
<Endpoint>1</Endpoint>
<ComponentType>FLAN</ComponentType>
<Weight>4.565</Weight>
<ComponentRefNo>=16386/43736</ComponentRefNo>
<OutsideDiameter>88.9000015258789</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<CorrosionAllowance>1</CorrosionAllowance>
<InsulationThickness>50</InsulationThickness>
<Position>-2000.00 13800.00 0.00</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
```

The first extra block comes after the SIF value and is a new Position marker.

```
<NewPosition>-2000 13800 0</NewPosition>
```

In most cases the new position matches exactly the position before because this value is only used if the actual position in the line has been changed.

Next comes the status line

```
<Status>True</Status>
```

Status can be:

'True' (no change)

'New' (a new node has been inserted)

'Deleted' (the node has been deleted)

'Changed' (some part of the definition has changed such as the type or loads on a restraint)

Deflections are catered for in lines after the status in load blocks. Each load case has a load block covering deflections for the current node:

```
<Load>
```

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

```
<LoadCaseName>CASE 1 (SUS) W</LoadCaseName>
<Translation>1.37 2.16 0.57</Translation>
<Rotation>0 0 0</Rotation>
</Load>
```

New items are added in the form of new nodes. The types of items to be added are normally restraints or bends and elbows for additional support or flexibility. Other new component types are not supported. The following is a new node to provide a restraint. Its status is New.

```
<Node>
<NodeNumber>55</NodeNumber>
<Endpoint />
<ComponentType>ATTA</ComponentType>
<Weight>0</Weight>
<ComponentRefNo />
<OutsideDiameter>88.9</OutsideDiameter>
<WallThickness>3.17</WallThickness>
<InsulationThickness>50</InsulationThickness>
<Position>-4551 13800 0</Position>
<BendRadius>0</BendRadius>
<SIF>0</SIF>
<Restraint>
<Type>+Z</Type>
<Stiffness>1751270000000</Stiffness>
<Gap>0</Gap>
<Friction>0</Friction>
</Restraint>
<NewPosition>-4551 13800 0</NewPosition>
<Status>New</Status>
<Load>
<LoadCaseName>CASE 1 (SUS) W</LoadCaseName>
<Translation>0 0 0</Translation>
<Rotation>0 0.0014 0</Rotation>
</Load>
</Node>
```

The return xml file is named after the original xml file but with the extension of RET.xml to indicate that it is a return file. It is a combination of the original and the new information from CAESAR and it is read into the model to produce a representation of the stress cases.

© 2015-2026 AVEVA Group Limited and its subsidiaries. All rights reserved.

AVEVA Group Limited
High Cross
Madingley Road
Cambridge
CB3 0HB
UK

Tel +44 (0)1223 556655

www.aveva.com

To find your local AVEVA office, visit
www.aveva.com/en/about/about-aveva/locations/

AVEVA believes the information in this publication is correct as of its publication date. As part of continued product development, such information is subject to change without prior notice and is related to the current software release. AVEVA is not responsible for any inadvertent errors. All product names mentioned are the trademarks of their respective holders.